

IF Multi-Factor Backtest Report

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1. Strategy Overview

Underlying: IF (CSI 300 Stock Index Futures), daily frequency
Predict Target: target_1d (next-day return)
Signal Filter: Enabled - flat when |prediction| < 0.5 x std
Short Threshold: 3.0x the long threshold (long-biased)
Models: Ridge Regression, Logistic Regression, XGBoost, LSTM

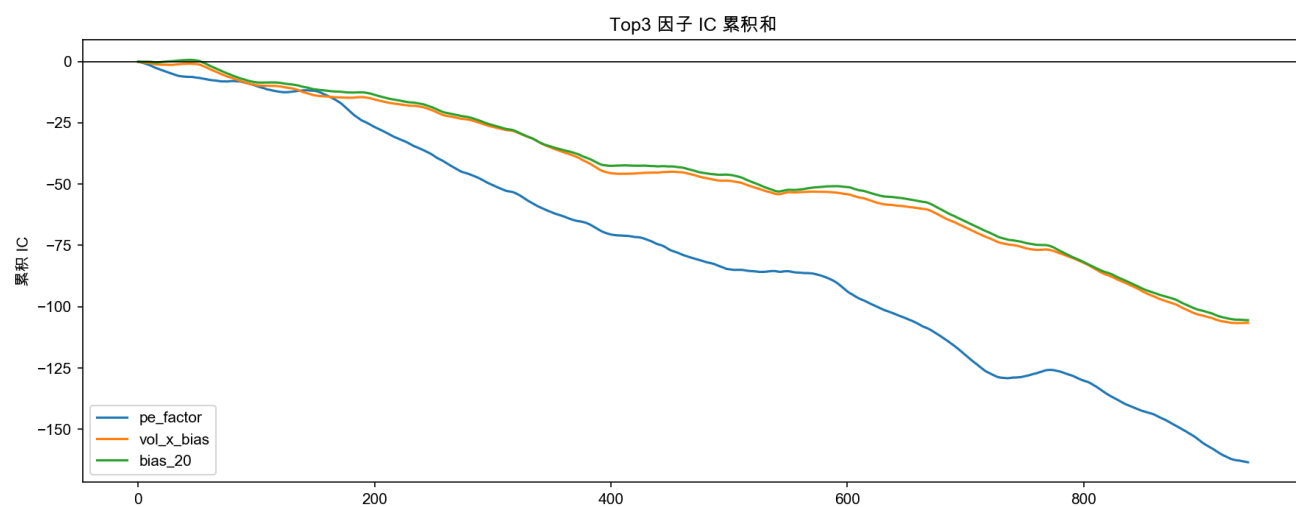
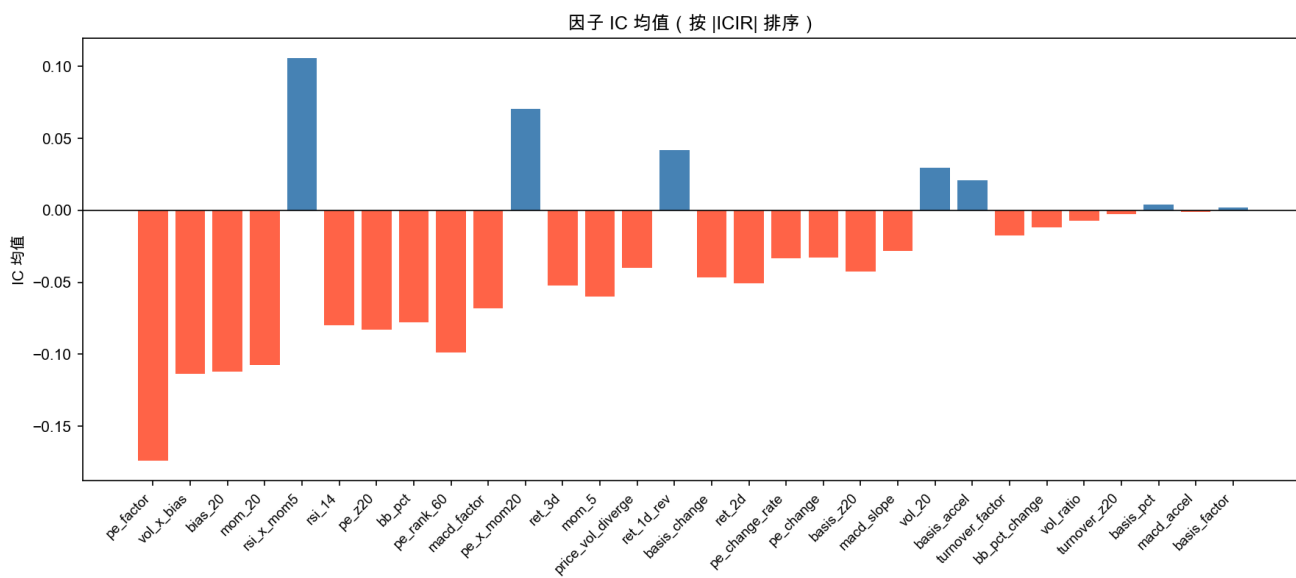
Core logic: construct price/volume and fundamental factors, screen by IC/ICIR validity, then feed selected factors into four ML models to predict return direction. Signals: long (+1) / short (-1) / flat (0). Train/test split: 70/30 by time.

2. Factor Analysis

Total factors constructed: 30 Selection criteria: |ICIR| >= 0.3, p < 0.05, pairwise correlation < 0.7
Selected factors (7): pe_factor, vol_x_bias, rsi_x_mom5, pe_x_mom20, ret_3d, price_vol_diverge, basis_change
Top 10 Factors by |ICIR|:

Factor	IC Mean	ICIR	IC>0 Rate	p-value
pe_factor	-0.1739	-1.4392	9.2%	< 0.0001
vol_x_bias	-0.1134	-1.2642	11.5%	< 0.0001
bias_20	-0.1123	-1.2400	12.9%	< 0.0001
mom_20	-0.1077	-1.0185	16.7%	< 0.0001
rsi_x_mom5	0.1057	0.8426	78.5%	< 0.0001
rsi_14	-0.0801	-0.8417	20.0%	< 0.0001
pe_z20	-0.0829	-0.8336	19.6%	< 0.0001
bb_pct	-0.0779	-0.8180	21.5%	< 0.0001
pe_rank_60	-0.0986	-0.7723	25.4%	< 0.0001
macd_factor	-0.0683	-0.7371	21.7%	< 0.0001

Note: Negative IC indicates a contrarian signal. Models learn the direction automatically.



3. Model Training Results

Time-based 70/30 train/test split. No random shuffle to prevent look-ahead bias.

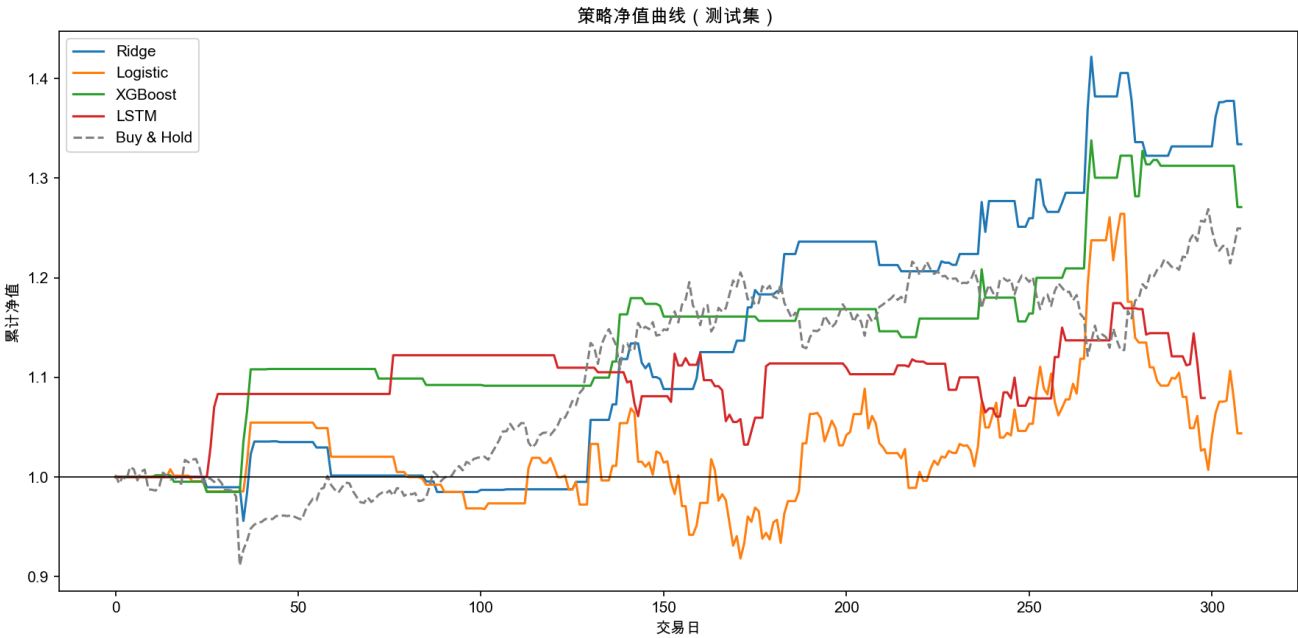
Best model selected by Sharpe ratio: Ridge

Model	Direction Acc.	IC	RMSE
Ridge	47.57%	0.0765	0.010919

4. Backtest Performance

Strategy return = signal x actual next-day return. Benchmark = Buy & Hold (long throughout the test period).

Model	Total Return	Ann. Return	Ann. Vol	Sharpe	Max DD	Win Rate
Ridge	33.40%	24.70%	15.56%	1.588	-6.99%	57.38%



5. Conclusion

Best Model: Ridge (Sharpe = 1.5878, Return = 33.40%)

Profitable Models: Ridge, Logistic, XGBoost, LSTM

Key Findings:

1. Most selected factors are contrarian (negative IC), indicating IF tends to mean-revert in the short term.
2. Non-linear models (LSTM) better capture complex factor-return relationships than linear models.
3. Signal filtering (flat zone) reduces low-confidence trades and limits drawdowns.
4. Single-asset time-series models have limited signal strength (IC mean $\sim 0.05-0.10$). Factor quality is the main bottleneck.

Disclaimer: Transaction costs (fees and slippage) are not included. Actual returns would be lower than reported.